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TREND-BASED MOCK PAPER

GATE Biotechnology

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Biotechnology
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Choose similar to 'catalyst':

- (A) Inhibitor
- (B) Accelerator
- (C) Precursor
- (D) Reactant

Section B: Biotechnology

Q2. [1-MCQ] [1 mark]

Normal distribution mean = median = ?

- (A) Mode
- (B) Median
- (C) Mean
- (D) SD

Q3. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q4. [1-NAT] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q6. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q7. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q8. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q9. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

- (A) $[1,0]^T$
- (B) $[2,1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q10. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q11. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q14. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q15. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x * \ln(x)$
- (C) $y = x * \sin(\ln x)$
- (D) $y = x^2$

Q16. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q17. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q20. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q21. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
 - (B) $(8/5, 4/5, 0)$
 - (C) $(2, 1, 0)$
 - (D) $(1, 2, 3)$
-

Q23. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q24. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q25. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q26. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q27. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q29. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q30. [1-NAT] [1 mark]

Photosynthesis light reactions produce:

(Numerical Answer Type — Enter the numerical value)

Q31. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA->DNA
 - (B) DNA->RNA
 - (C) Protein->DNA
 - (D) RNA->Protein
-

Q32. [1-MCQ] [1 mark]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
 - (B) Nucleus
 - (C) Mitochondria only
 - (D) G1 phase
-

Q33. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
 - (B) 9:3:3:1
 - (C) 1:2:1
 - (D) 1:1:1:1
-

Q34. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
 - (B) Decreases
 - (C) Unchanged
 - (D) Zero
-

Q35. [2-NAT] [2 marks]

Central dogma: reverse transcription is:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-NAT] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-NAT] [2 marks]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MSQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q39. [2-MSQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q40. [2-MSQ] [2 marks]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
- (B) Nucleus
- (C) Mitochondria only
- (D) G1 phase

Q41. [2-MCQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q42. [1-MCQ] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q43. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q44. [1-NAT] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q45. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q46. [1-NAT] [1 mark]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-NAT] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
 - (B) 0.693 min
 - (C) 1.44 min
 - (D) 0.5 min
-

Q49. [2-NAT] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Strongest acid:

- (A) HCl
 - (B) HNO₃
 - (C) H₂SO₄
 - (D) HClO₄
-

Q51. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
 - (B) Chemical to electrical
 - (C) Heat to work
 - (D) Mechanical to electrical
-

Q52. [1-MCQ] [1 mark]

Law of segregation describes:

- (A) Alleles
 - (B) Genes
 - (C) Chromosomes
 - (D) DNA
-

Q53. [1-MCQ] [1 mark]

DNA stands for:

- (A) Deoxyribonucleic Acid
 - (B) Ribonucleic
 - (C) Dinitrogen Acid
 - (D) None
-

Q54. [1-MCQ] [1 mark]

DNA B-form base pairs per turn:

- (A) 10

- (B) 10.4
- (C) 12
- (D) 8

Q55. [1-NAT] [1 mark]

Linked genes show:

(Numerical Answer Type — Enter the numerical value)

Q56. [1-NAT] [1 mark]

ABO blood group has how many alleles?

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q58. [2-MCQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q59. [2-MCQ] [2 marks]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q60. [1-MCQ] [1 mark]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q61. [1-MCQ] [1 mark]

Binary fission in:

- (A) Bacteria
- (B) Fungi
- (C) Plants
- (D) Animals

Q62. [1-MCQ] [1 mark]

Batch process is:

- (A) Continuous
- (B) Intermittent

- (C) Steady
- (D) None

Q63. [1-MCQ] [1 mark]

Antibody produced by:

- (A) T cells
- (B) B cells
- (C) Macrophages
- (D) None

Q64. [1-NAT] [1 mark]

BLAST is for:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Chromosome number in daughter:

- (A) Halved
 - (B) Same
 - (C) Doubled
 - (D) Variable
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	C	1
3	1MCQ	A	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1NAT	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	A	1
13	1MCQ	B	1

14	1MCQ	A	1
15	1NAT	B	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1NAT	B	1
19	1MCQ	B	1
20	1NAT	C	1
21	1NAT	B	1
22	1MCQ	C	1
23	1MCQ	C	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1NAT	B	1
28	1MCQ	D	1

29	1MCQ	A	1
30	1NAT	D	1
31	2MCQ	B	2
32	2MCQ	B	2
33	2NAT	B	2
34	2NAT	A	2
35	2MSQ	C	2
36	2MCQ	A	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2NAT	A	2
40	2MCQ	A	2
41	2NAT	C	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2NAT	B	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	B	2
49	2MCQ	A	2
50	2NAT	A	2
51	2NAT	A	2
52	2NAT	B	2
53	2NAT	B	2
54	2MCQ	A	2
55	2NAT	B	2
56	2MSQ	B	2
57	2MCQ	B	2
58	2MCQ	A	2

59	2MSQ	A	2
60	2MSQ	B	2
61	2NAT	B	2
62	2NAT	B	2
63	2MCQ	A	2
64	2MCQ	D	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Mendel | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Molecular Biology | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Biochemistry | Topic: Proteins | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Microbiology | Topic: Bacteria | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Bioprocess Engineering | Topic: Fermentation | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Immunology | Topic: Antibodies | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Bioinformatics | Topic: Sequence | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitosis | Difficulty: moderate

Q11.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q12.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q13.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q14.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q16.

Answer: A

Solution:

Reverse transcription: RNA → DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q17.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q18.

Answer: B

Solution:

$\Delta_n = 2 - 4 = -2.$

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q19.

Answer: B

Solution:

AaBb × AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q20.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q21.

Answer: B

Solution:

|x| is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q25.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

ABO: I^A , I^B , i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q28.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q29.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}} \approx 1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q31.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q32.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

$AaBb \times aabb$: offspring $AaBb$, $Aabb$, $aaBb$, $aabb$ in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q41.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q43.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q46.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: A

Solution:

$$L\{t^*f(t)\} = -d/ds(F(s)). \quad F(s) = 1/(s+2). \quad -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q52.

Answer: B

Solution:

$$\Delta_n = 2 - 4 = -2.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q53.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q54.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q55.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q56.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q57.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q58.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q59.

Answer: A

Solution:

Reverse transcription: RNA->DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q60.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q61.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2.$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q62.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: A

Solution:

Reverse transcription: RNA->DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q64.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q65.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 28 questions, 48 marks
Biology: 12 questions, 19 marks
Chemistry: 10 questions, 15 marks
Genetics: 8 questions, 11 marks
General Aptitude: 1 questions, 1 marks
Biochemistry: 1 questions, 1 marks
Microbiology: 1 questions, 1 marks
Bioprocess Engineering: 1 questions, 1 marks
Immunology: 1 questions, 1 marks
Bioinformatics: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 18 | Moderate: 40 | Difficult: 7

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.